

Knot Mutations and Khovanov Homology

A Thesis

Submitted to the Graduate Faculty of the
University of South Alabama
in partial fulfillment of the
requirements for the degree of

Master of Science

in

Mathematics

by

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B.S., University of California at San Diego, 2018
August 2020

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This thesis is dedicated to Sophia Hoang and the pbros who housed me so I could
get some necessary R&R during my San Diego getaways.

ACKNOWLEDGEMENTS

I would first like to give my greatest gratitude to my thesis advisor Dr. Christine Lee. Since I started my master's program, you have consistently guided me and given me the opportunity to showcase my knowledge and growth as an individual. Without your constant support, encouragement, friendliness, and patience, I would not have made it to where I am today. There were many times I struggled to adjust to life in Alabama and get past these long two years, but your encouragement allowed me to push through my self-doubts and finish this program.

My sincere thanks also goes to my committee members: Dr. Carter, Dr. Godang, and Dr. Pillen. To Dr. Carter, thank you for taking the time to provide valuable feedback to my thesis, even during your retirement! I hope one day we will be able to officially meet in person. To Dr. Godang, thank you for taking the time to provide valuable advice to my thesis. Your friendliness and positivity during our small interactions in the hallways of the mathematics building helped make me feel more at home. And to Dr. Pillen, thank you as well for your advice on my thesis, for being a great Linear and Abstract Algebra professor and an amazing graduate coordinator. I have learned a lot from you and both of your courses, and I thoroughly enjoyed your classes and our seminar.

I would also like to say thank you to the McLaurins for their fellowship. Without your help, I would not have been able to progress through my thesis during the summer in a timely and efficient manner.

A special shout-out to the other math graduate students in the department that I have had the pleasure of getting to know and become friends with over the last two years at this university. You guys made my master's program bearable with all of the camaraderie in the graduate student offices.

And last, but not least, I would like to thank my family for the encouragement and support for the time I was away to work on my master's program. Your emotional and financial support allowed me to study here in Alabama as comfortable as possible, especially being so far away from home.

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ABSTRACT

Jonathan R. Tin, M.S., University of South Alabama, August 2020, Knot Mutations and Khovanov Homology, Chair of Committee: Christine Ruey Shan Lee, Ph.D.

This thesis will determine whether or not Khovanov homology can detect knot mutation. Khovanov homology is a knot invariant introduced by Mikhail Khovanov in the late 1990s. Before addressing the main problem, the thesis outlines some fundamentals of knot theory, including the definition of Khovanov homology. Finally, utilizing this invariant we investigate a specific knot and its mutation.

CHAPTER I

INTRODUCTION

Around the end of the 19th century, knot theory started to gain much traction in being its own mathematical discipline. Physicsts Tait and Thomson initial research with smoke rings garnered much inspiration for knot theory. Tait's (debunked) theory that atoms were small bodies of knotted vertices in the *ether* led to the desire to classify and catalogue knots. Additionally, Gauss during this time had played around with the idea of knots, which then inspired his student, Listing, to take his ideas further and study what is essentially knot theory: the mathematical examination of embeddings of simple closed curves in three-dimensional space. Listing, is credited to coining the term “topologie” in 1836 and the exploration of topological inquiries to knots [7][22]. Profound progress was made in the 20th century towards these studies. A few noteworthy mathematicians during this time are Alexander, Reidemeister, and Wirtinger who developed algebraic quantities that could be used to distinguish knots. Such a quantity is called an *invariant* in knot theory, because it is defined using some specific (diagrammatic) description, but it remains unchanged under such description, or under a topological deformation.

Following Alexander's work on polynomial invariants (which is now regarded as the ‘Alexander Polynomial’), in [8] Conway created key ideas. He described the Alexander polynomial recursively via a *skein relation*. He developed a theory of tangles and demonstrated that tangles could undergo a *mutation*, yet their associated skein polynomials would remain unchanged. The word “*skein*”, defined by Merriam-

Webster dictionary, is a loosely coiled length of yarn or thread wound on a reel. Conway's version of a skein polynomial eventually evolved through other considerations into the Jones Polynomial [13]. This, in turn, grew into Khovanov homology, in the sense that the Jones polynomial is a graded Euler characteristic of the Khovanov homology groups. In this thesis, we give two mutant knots that have different Khovanov homology groups. Interestingly enough, Khovanov homology is a stronger invariant than the Jones polynomial, even though Khovanov homology arises from skein theory. This makes studying the behavior of Khovanov homology under mutation quite intriguing.

A first goal of this thesis is to present the ideas involved in knot theory. We discuss the idea of knot mutation and motivate and introduce several foundational techniques, i.e. knot invariants, used to help distinguish knots. The next major goal of this thesis is to apply what we have discussed about knot mutations and knot invariants, and explore Khovanov homology. We will use this knot invariant to help us determine if this is able to detect knot mutations by applying it to our two knot mutant knots.

To this end, the fundamentals of knot theory are introduced in Chapter II. This chapter contains relevant information that are pertinent to our discussion including: an explicit description of knots, their particular properties, and certain knot invariants. Chapter III contains the motivation behind this thesis by examining knot mutations and the Jones polynomial. Finally in Chapter IV Khovanov homology is introduced and applied to our problem at hand. In Chapter V, we summarize our progress provided in this thesis. A major question worth exploring is if it is possible for Khovanov homology to distinguish mutant knots, and we hope to provide more insight to this.

CHAPTER II

KNOTS AND KNOT INVARIANTS

2.1 Knots and Links

The idea of a knot is an infinitely flexible loop of rubber that is embedded in 3-space that is considered up to ambient isotopy (which we will discuss more formally later). Precisely, a knot is a one component link, and a link is defined as follows:

Definition 2.1. A *link* L of n components is a subset of $\mathbb{S}^3$ that consists of n disjoint, piecewise linear, simple closed curves. That is a link is an embedding of the union of finitely many (flexible) circles in $\mathbb{S}^3$.

Definition 2.2. A *smooth knot* is a smooth embedding of the circle, $\mathbb{S}^1$, into the 3-sphere, $\mathbb{S}^3$, that is considered up to smooth isotopy (we informally discuss this more in Remark 2.10).

Example 2.3. Below are some examples of knots.

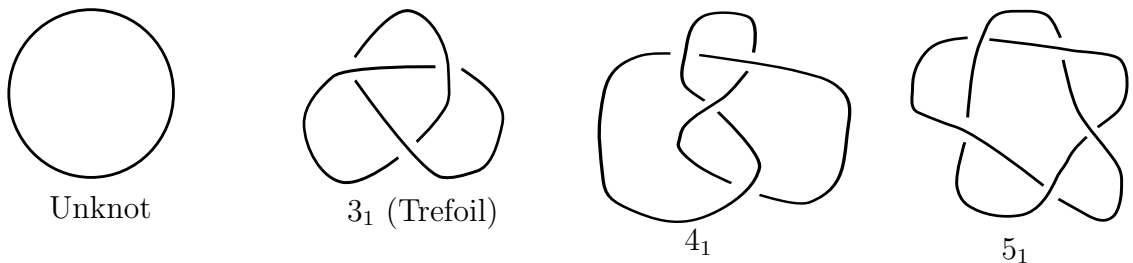


Figure 1. Knot Examples.

Example 2.4. Below are some examples of links.

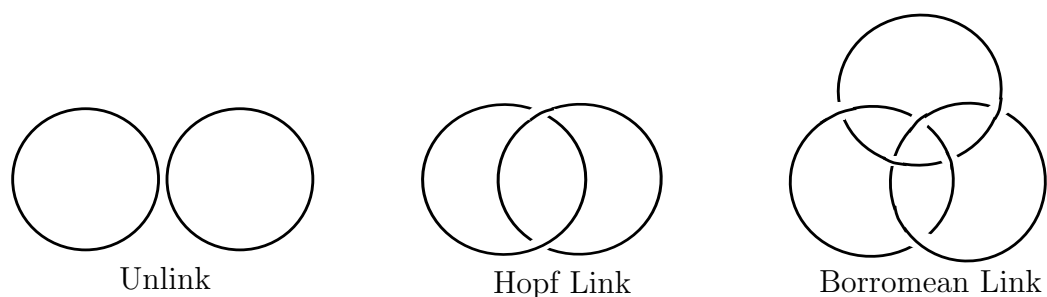


Figure 2. Link Examples.

Before continuing on, it is important to note that Figure 1 and Figure 2 are actually diagrams of knots and links, which is what we will always be working with. This is formally discussed in Definition 2.13.

2.2 Knot Equivalency and Isotopy

Now we discuss when we consider two knots to be the *same* and when they are different. For example, consider these two knots below:

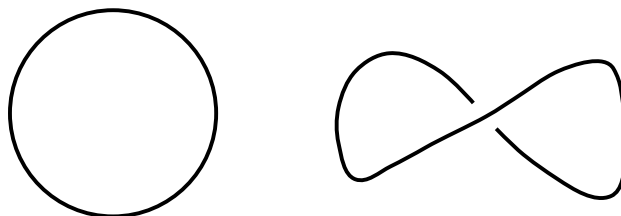


Figure 3. Two Knots Which Look Similar.

The knot on the left is the unknot, and the one on the right is some other knot that looks quite like the unknot. We would like to consider these two knots to be the same because we can simply wiggle the knot on the right at its center and produce the unknot from this wiggling.

Definition 2.5. An *isotopy* is a fibrewise-continuous mapping $f : X \times I \rightarrow Y \times I$ such that f takes the fibre $X \times t$ homeomorphically onto a subset of the fibre $Y \times t$. In other words, an isotopy $F_t : Y \rightarrow Y$, for which $F_t(Y) = Y$ for each t , is called an isotopy of the space Y [21].

Definition 2.6. Two knots K_1 and K_2 in $\mathbb{S}^3$ are *ambient isotopic* if there exists an isotopy $H : \mathbb{S}^3 \times [0, 1] \rightarrow \mathbb{S}^3$ such that $H(K_1, 0) = K_1$ and $H(K_1, 1) = K_2$.

Definition 2.7. We define two knots to be *equivalent* if they are related to each other by an *ambient isotopy*.

Fact 2.8. Burde proved that the smooth knot and the piecewise linear simple closed curve are the same [6].

Definition 2.9. A *knot* K is a one component link.

Remark 2.10. To expand a little more from Remark 2.2 on smooth knots and smooth isotopies with respect to $\mathbb{S}^3$, *smooth isotopy* is a smooth homotopy between two smooth knots such that they are equivalent if there exists an orientation preserving diffeomorphism $\mathbb{S}^3 \rightarrow \mathbb{S}^3$, taking one embedding to the other. Smooth isotopy classes of knots are the same as regular isotopy classes of knots, as we have defined earlier.

2.3 Knot Projections and Knot Diagrams

A thought that might occur is that there are potentially many possibilities of knots that are considered to be the same, but just look different. This leads to the question:

How exactly can we go about finding out ways to distinguish these knots? In order to do this, there have been various *knot invariants* developed - methods that can tell us which knots are inequivalent to each other.

For our purposes, we do not work with the actual knot itself, but rather a diagram of the knot. To get a diagram of a knot, we project our knot from $\mathbb{S}^3 \rightarrow \mathbb{S}^2$, which will give rise to a knot diagram D . By projecting the knot, we are taking a surjective map from $\mathbb{S}^3 \rightarrow \mathbb{S}^2$, and we consider the image of the knot under that map. Diagrams are general position projections, which only transverse up to double points and has no sharp bends. More formally,

Definition 2.11. A *knot diagram* D is a projection of a knot from $\mathbb{S}^3 \rightarrow \mathbb{S}^2$, where we have no more than 2 points that are superposed to each other. We can then draw the diagram with a “gap”, which represents the superposed points on $\mathbb{S}^3$, and draw in the information of which crossings are over/under.

We call these projections that fit our definition of a knot diagram *regular projections*. If a projection does not meet this double point restriction, and also projections that have tangents and cusps or sharp points, then we will call these projections *irregular projections*.

Remark 2.12. As such, throughout the rest of the thesis, assume all projections are regular.

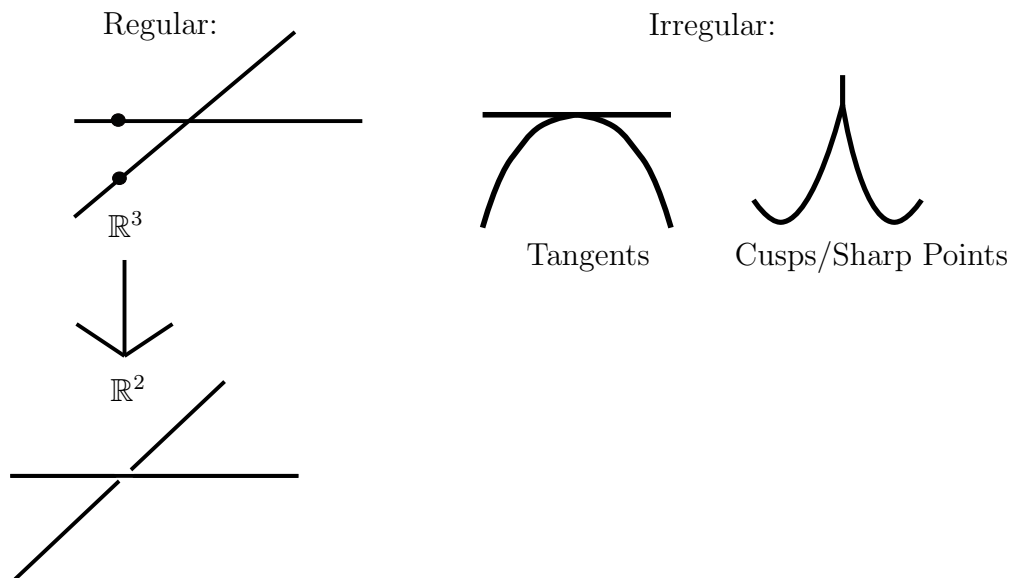


Figure 4. Regular and Irregular Projections.

Now that we have our knot diagram D , we can also assign an orientation to its diagram as well.

Definition 2.13. An *oriented knot diagram* D , is simply a knot diagram with a chosen direction that circulates the entire diagram (we denote this with an arrow on the diagram itself).

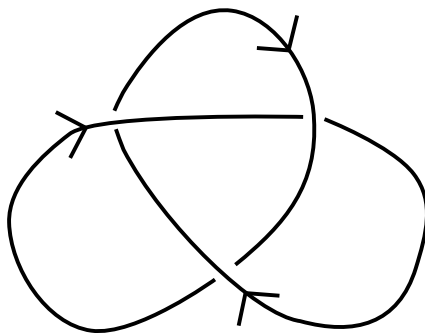


Figure 5. Oriented Diagram of the Trefoil Knot.

2.4 Reidemeister Moves and Knot Invariants

To further discuss ambient isotopy on the knot, Reidemeister described what are called Δ -moves. This will allow us to define our equivalence relation on knots in regards to this deformation, i.e. an equivalence relation is achieved through a sequence of Δ -moves and its reverse, Δ' -moves. He demonstrated that these are sufficient, and that they reduce to three types, as mentioned in Theorem 2.18. To demonstrate this clearly, we will represent our knot as polygonal (disregarding Fact 2.8 for this part).

Definition 2.14. Let K be a knot with $[A, B]$ as one of its edges and where a triangle, $[A, B, C]$, with C not in K , does not intersect K . To perform a Δ -move we can “remove” $[A, B]$ by “adding” in two edges, $[A, C]$ and $[C, B]$. The reverse move, Δ' , of this holds as well. See top of Figure 6.

Remark 2.15. We are also allowed to *split* an edge, say $[A, B]$, into two edges by placing a point C on $[A, B]$, such that we now have adjacent edges $[A, C]$ and $[C, B]$. And the converse of this holds as well, where we we can remove point C to just get $[A, B]$. See bottom of Figure 6.

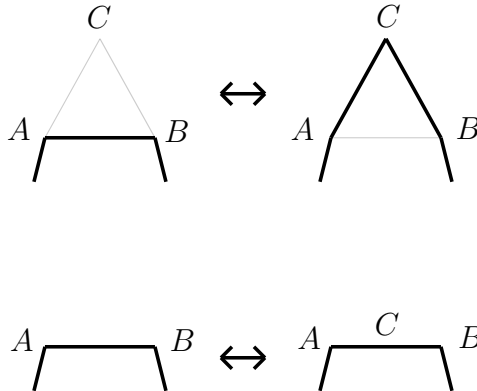


Figure 6. Δ -move.

Theorem 2.16. *Two knot diagrams are equivalent to each other if they can be related by a sequence of Δ and Δ' moves.*

Remark 2.17. The converse of Theorem 2.16 is true as well [19].

Because we can perform any number of these Δ -moves on the knot itself to get an equivalent knot, we have to carefully consider what happens to its diagram. Notice that a given knot can be represented by many diagrams, none of which will be different knots as they differ by Δ -moves. This causes a problem since if any knot can be represented by any number of diagrams, it can be rather unclear how much accurate information (such as its minimum number of crossings) we can truly extract from a knot diagram. For instance, looking at the two equivalent knots in Figure 3, which differ by a single Δ -move, how can we know for certain that their individual projections are also equivalent? We now address how to extract information about a knot from a representative diagram by means of Reidemeister's theorem, which he proved in the 1930's, showing that these diagrams are indeed equivalent up to ambient isotopy [18].

Theorem 2.18. (*Reidemeister's Theorem*): *Two knots are equivalent if and only if their diagrams are related to each other by a finite sequence of Reidemeister Moves [18].*

There are three Reidemeister moves which are:

1. RI (Move I): We twist and untwist in either direction.
2. RII (Move II): We move one loop completely over another.
3. RIII (Move III): We move a string completely over or under a crossing.

Remark 2.19. Reidemeister proved his theorem with the aforementioned Δ -moves discussed in Definition 2.14 [18].

Remark 2.20. It is worthwhile to mention that Alexander and Briggs had independently proved this as well, approximately 6 months after Reidemeister [2].

Remark 2.21. When using Reidemeister moves, a useful technique to employ at times when working with knot diagrams is *planar isotopy*. This comes from the equivalence relation that is generated in our Δ -moves defined earlier, but acts in $\mathbb{S}^2$. So planar isotopy is just the deformation of the knot projection, without changing the actual knots structure (like its crossings).

Example 2.22. Below are diagrams depicting each of the three moves:

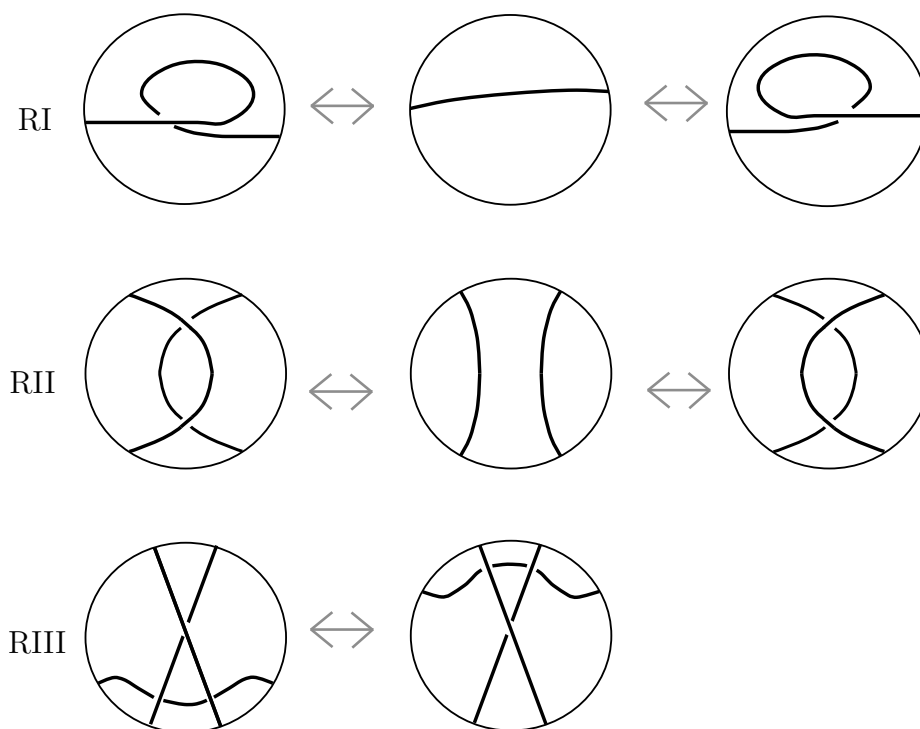


Figure 7. Reidemeister's Moves.

Even with such a powerful result proven by Reidemeister, it still has its limits in distinguishing knots. As noted earlier, since a knot can be represented by many different diagrams, and each diagram is a specific representation of it, we want to be able to find the quantities that remain unchanged when we have different diagrams representing the same knot. Thus we can strengthen Reidemeister's moves further by creating knot invariants with them, in turn creating better methods to distinguish knots through its various diagrams.

Definition 2.23. A *knot invariant* is some information, i.e. a quantity, such as a number or polynomial which remains the same under isotopy of knots.

Remark 2.24. In Chapter 4, an algebraic quantity, Khovanov homology, is used to distinguish a pair of mutant knots. It is defined via representative diagrams. Khovanov proves that it remains unchanged under the Reidemeister moves [16].

Now it is known that if two knots have different invariants then they are inequivalent knots. Because of this, knot invariants give us a great way of distinguishing inequivalent knots. However it is not necessarily true that because two knots share the same invariant, that they are equivalent knots, as we will see in Example 3.16 later on. In particular, the Jones polynomial, discussed next, is a polynomial invariant that distinguishes knots that were previously known to be difficult to distinguish.

CHAPTER III

THE JONES POLYNOMIAL AND MUTATION

3.1 Kauffman Bracket

The Jones polynomial was discovered by Vaughan Jones in 1984 [13]. It is a powerful invariant for oriented knots, which assigns a Laurent polynomial to the oriented knot diagram. Then in 1987, Louis Kauffman discovered a state-sum model via his own definition of the Jones polynomial, the Kauffman bracket, which could be used to calculate the Jones polynomial as well [14]. In defining his polynomial, Kauffman provided a modification to Conway's skein relation. To show the construction of the Jones polynomial, we will present Kauffman's definition of the Jones polynomial. Kauffman's construction is more useful for us because it requires less mathematical machinery to understand. Additionally, its bracket relation is used to construct Khovanov homology, and its tree of computations can be arranged in a *hyper-cube*, making it ideal for us to examine the Jones polynomial from this viewpoint. This arrangement suggests Khovanov homology, which will be discussed in Chapter IV.

Definition 3.1. Given an unoriented knot diagram D , the *Kauffman bracket polynomial*, $\langle D \rangle$, is defined on the ring $\mathbb{Z}[A^{\pm 1}]$ by the following rules:

1. It is calculated using the *skein relation*, depicted in Figure 8, where for a crossing of D we can assign any α smoothing to have weight A and any β smoothing to have weight A^{-1} .

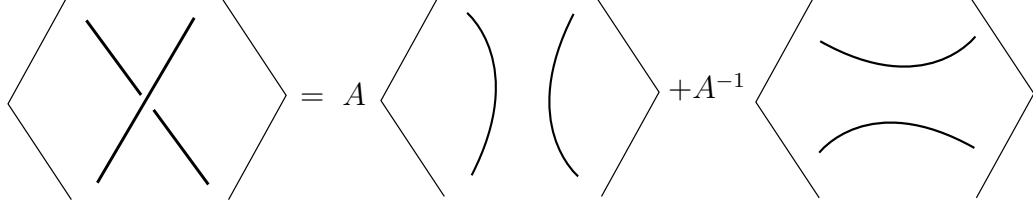


Figure 8. Left Smoothing: α and Right Smoothing: β .

2. It satisfies $\langle O \cup D \rangle = (-A^2 - A^{-2})\langle D \rangle$, where O is any diagram of the unknot.
This means that we can remove any unknot diagrams at the cost of multiplying our polynomial by $(-A^2 - A^{-2})$.
3. The normalization $\langle O \rangle = 1$. So the bracket of any unknot diagram is equal to 1.

Example 3.2. We will do a simple example to illustrate Definition 3.1. In (3) of Definition 3.1, the normalization tells us the value of the unknot:

Figure 9. Kauffman Bracket Normalization.

And if we have a diagram of an unlink, by (2) we get that $(-A^2 - A^{-2}) \cdot 1$ (i.e. times the bracket of the unknot diagram):

$$\left\langle \bigcirc \bigcirc \right\rangle = (-A^2 - A^{-2})$$

Figure 10. Kauffman Bracket of Unlink.

So to calculate a 1 crossing unknot diagram, we apply (1) and get:

$$\left\langle \text{figure-eight} \right\rangle = A \left\langle \bigcirc \bigcirc \right\rangle + A^{-1} \left\langle \text{planar} \right\rangle$$

Figure 11. Kauffman Bracket 1 Crossing Unknot.

The first diagram we already calculated above. The second diagram is planar isotopic to the unknot; so, by (3) it is just 1. Altogether now we have $A(-A^2 - A^{-2}) + A^{-1} = A^3$.

Now a method to calculate the Kauffman bracket of a knot diagram, is to take a *state sum* over all all possible states. To find these states, we resolve each crossing one at a time by applying both smoothings, and then sum over all possible states with its corresponding weight as mentioned in Definition 3.1.

Definition 3.3. A *state* of a knot diagram D is the result of applying a defined set of smoothings to the diagram.

Remark 3.4. Recall from Definition 3.1 that an α smoothing is assigned a weight of A and a β smoothing is assigned a weight of A^{-1} . So for each crossing we resolve, there are two possible smoothings; thus, we will obtain two states, with weight A and A^{-1} , for each resolved crossing.

Proposition 3.5. *The Kauffman bracket can be calculated by*

$$\langle D \rangle = \sum_S \langle D|S \rangle d^{|S|-1}$$

where $\langle D|S \rangle d^{|S|-1} = A \sum_s (-A^2 - A^{-2})^{|S|-1}$ (the total contribution of a given state to the polynomial), $\sum_S$ is the summation over all states of D , and $|S|$ is the number of circles for each corresponding state.

Example 3.6. Looking at Figure 12, after applying the appropriate α and β smoothings to the top and then bottom crossings, we can set up the following hyper-cube of states with its appropriate weight, and construct a sum of all of its states (i.e. a state-sum):

$$\begin{aligned} & A^2(-A^2 - A^{-2}) + 1 \cdot 1 + 1 \cdot 1 + A^{-2}(-A^2 - A^{-2}) \\ &= -A^4 - A^0 + 1 + 1 - A^0 - A^{-4} \\ &= -A^4 - A^{-4} \end{aligned}$$

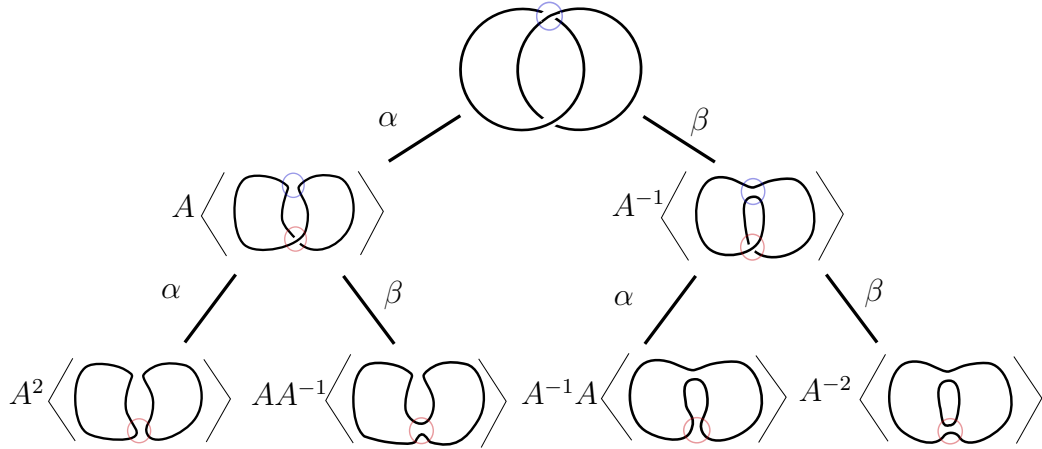


Figure 12. Kauffman Bracket Example.

Remark 3.7. Notice that from this calculation we can see that the Kauffman bracket is indeed well-defined, as it would not matter which crossing we resolve first.

Now as it turns out, the way the Kauffman bracket is currently constructed, up to this point, is almost a knot invariant with a slight caveat.

Lemma 3.8. *The Kauffman bracket is not invariant under Reidemeister move I.*

Proof. To see this, we can easily compute the following:

$$\begin{aligned}
\text{(i)} \quad \left\langle \text{Diagram 1} \right\rangle &= A \left\langle \text{Diagram 2} \right\rangle + A^{-1} \left\langle \text{Diagram 3} \right\rangle \\
&= A \cdot 1 + A^{-1}(-A^2 - A^{-2}) \cdot \left\langle \text{Diagram 4} \right\rangle \\
&= (A - A - A^{-3}) \cdot \left\langle \text{Diagram 4} \right\rangle \\
&= (-A^{-3}) \cdot \left\langle \text{Diagram 4} \right\rangle \\
\\
\text{(ii)} \quad \left\langle \text{Diagram 1} \right\rangle &= A \left\langle \text{Diagram 3} \right\rangle + A^{-1} \left\langle \text{Diagram 2} \right\rangle \\
&= A \cdot (-A^2 - A^{-2}) \cdot \left\langle \text{Diagram 4} \right\rangle + A^{-1} \cdot 1 \\
&= (-A^3 - A^{-1} + A^{-1}) \cdot \left\langle \text{Diagram 4} \right\rangle \\
&= (-A^3) \cdot \left\langle \text{Diagram 4} \right\rangle
\end{aligned}$$

Figure 13. R. Move I is non-invariant.

As you can see, we get two different values under the Reidemeister move I. $\square$

Lemma 3.9. *The Kauffman bracket is invariant under Reidemeister move II and III.*

A proof of these can be found in Kauffman's paper [14].

Now that we see the Kauffman bracket failing to be an invariant since it cannot distinguish Reidemeister move I, we will introduce the *writhe* of an oriented knot diagram to remedy this.

Definition 3.10. Given an oriented knot diagram D , its *writhe*, $Wr(D)$, is the sum of the signs of all the crossings (where n is the number of crossings) of D . Positive crossings, n_+ are given a value of $+1$, and negative crossings, n_- , are given a value of -1 i.e. $Wr(D) = \text{"number of" } n_+ \text{ crossings} - \text{"number of" } n_- \text{ crossings}$.

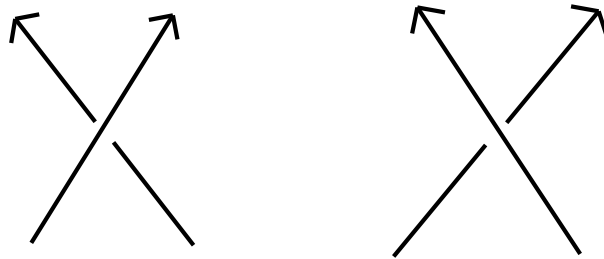


Figure 14. Positive and Negative Crossing.

As it turns out the writhe will either increase or decrease by 1 under a Reidemeister move I, but is invariant under moves II and III. Thus as long as we take orientation into account, we can get the Kauffman bracket polynomial to be a knot invariant.

Theorem 3.11. *Given an oriented knot diagram D , the Laurent polynomial $f(D) = (-A)^{-3Wr(D)}\langle D \rangle$, is invariant under the Reidemeister moves.*

Proof. We will show that the Kauffman bracket is invariant under a Reidemeister move I. For Reidemeister moves of type II and III we refer again to [14]. Looking at what happens with Reidemeister move I (as seen in Figure 13), notice that if we add in a positive crossing to D , we increase the writhe by 1 and multiply the Kauffman bracket by $-A^3$. This will not change $f(D)$. Now similarly, if we add in a negative crossing to D , the writhe decreases by 1 and multiply the Kauffman bracket by $-A^{-3}$. Thus we can define $f(D) = (-A)^{-3Wr(D)}\langle D \rangle$, which remains unchanged under Reidemeister move I. $\square$

Remark 3.12. An interesting fact that Kauffman proves in his paper, is that the Kauffman bracket can distinguish mirror images of knot diagrams [14].

Kauffman proved that his bracket can define the Jones polynomial.

Theorem 3.13. *The Jones polynomial $V(K)$, where K is an oriented knot diagram, can be obtained by computing $f(D) = (-A)^{-3Wr(D)}\langle D \rangle$, and then making a change of variable of $A = t^{-\frac{1}{4}}$ [14].*

Example 3.14. From Example 3.6, we found the Kauffman bracket polynomial of the Hopf link to be $-A^4 - A^{-4}$. Thus, if we assign each circle of the hopf link in Figure 12 a clockwise orientation, we can find $f(D)$ and make our change of variables to get the Jones polynomial of the hopf link: $V(L) = -t - t^{-1}$.

3.2 Mutant Knots

In this section, we show that the Jones polynomial is invariant under mutation. This fact is one of the main motivations behind the thesis.

Definition 3.15. Given a knot diagram D , a *mutation* to D gives rise to a different diagram D' by performing one of three moves:

1. A horizontal rotation
2. A vertical rotation
3. A 180° rotation

Note that this does not necessarily give rise to a new knot.

In Figure 15, we draw a boundary circle, where the inside represents a part of the knot diagram, such that 4 strands emerge from it. We then perform a mutation on the boundary circle such that the parts inside the circle are changed, but everything outside of it remains the same. In other words, all the strands of the diagram remain unchanged outside the boundary circle.

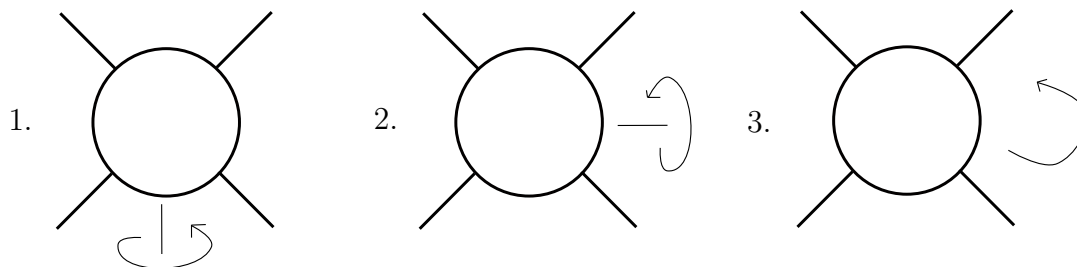


Figure 15. The Three Mutation Moves.

Example 3.16. A famous example of a knot and its mutant, which are actually different knots but have the same Jones polynomial, are the Kinoshita-Terasaka and Conway Knot, as seen in Figure 16. It was first proven by Riley in 1971 that they are in fact different knots [20].

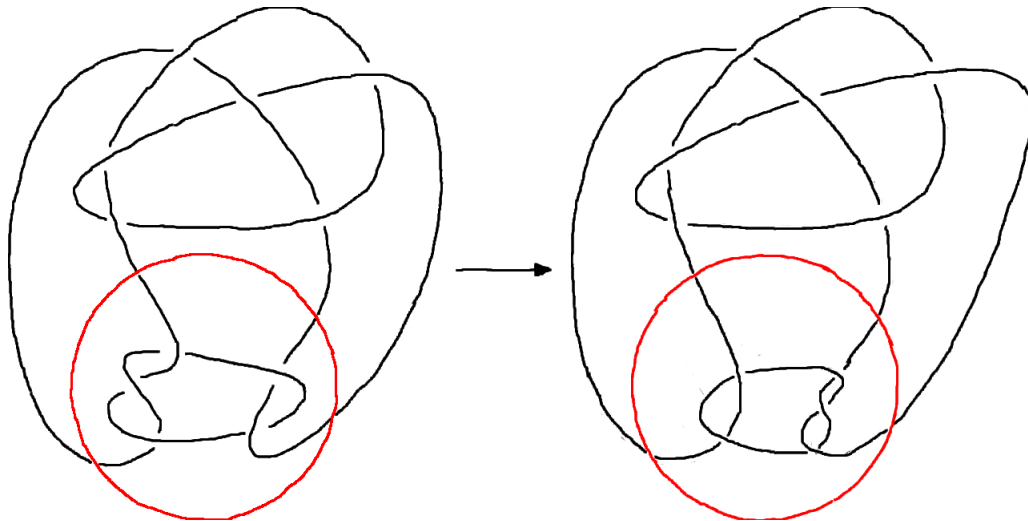


Figure 16. The left is Kinoshita-Terasaka and the right is Conway Knot, by a horizontal rotation.

3.3 Exploring Other Invariants

In this section, we will see why the Jones polynomial cannot distinguish mutant knots.

Remark 3.17. The Jones polynomial cannot distinguish different knots which are related by a mutation.

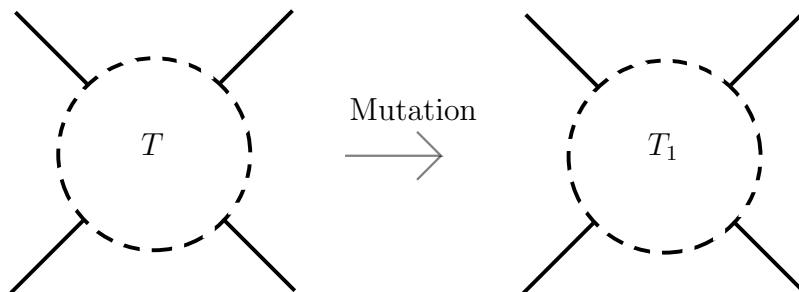


Figure 17. Sub-diagram T and T_1 with 4 Strands Emerging from it.

Proof. First we show that the Kauffman bracket is unchanged under mutation. So let T and T_1 , which are related by one of the three mutation moves, be a sub-diagram as seen in Figure 17.

Let $\langle D \rangle$ be the diagram and C a circle containing T . Similarly, let $\langle D' \rangle$ be a new diagram, where C contains T_1 , which has been rotated by one of the mutation moves. Depending on the diagram, by performing a mutation, the horizontal or vertical rotation will either keep the crossings the same, or it will flip them (i.e. over-under because under-over or vice-versa). However, due to our skein relation and that we are not considering orientation, this would not affect how we derive our α and β smoothings. For the 180° rotation, both α and β smoothings are invariant under this. So, in fact, $\langle D' \rangle$ covers all of the mutation cases.

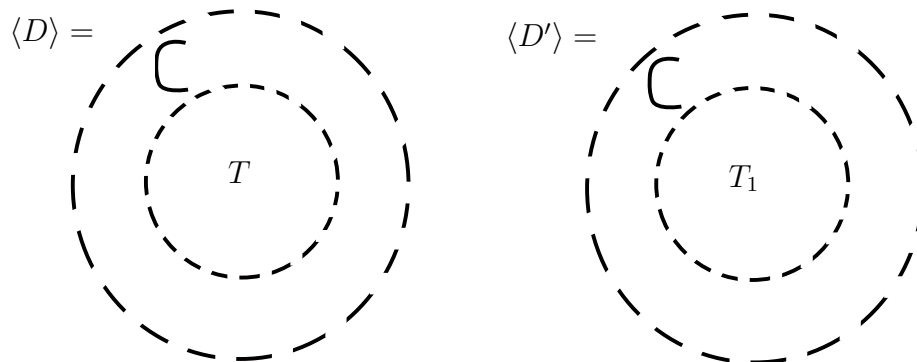


Figure 18. Figure for $\langle D \rangle$ and $\langle D' \rangle$.

Using the idea of the Kauffman bracket and skein relation, we will resolve all crossings in T and T_1 , and any unknots $= 1$, so we get:

$$\begin{aligned}
& \left\langle \begin{array}{c} \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \end{array} \right\rangle = F(A) \left\langle \begin{array}{c} \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \end{array} \right\rangle + G(A) \left\langle \begin{array}{c} \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \end{array} \right\rangle \\
& \left\langle \begin{array}{c} \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \end{array} \right\rangle = F(A) \left\langle \begin{array}{c} \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \end{array} \right\rangle + G(A) \left\langle \begin{array}{c} \diagup \quad \diagdown \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \diagdown \quad \diagup \end{array} \right\rangle
\end{aligned}$$

Figure 19. Proof of Crossings Remaining Unchanged Under Mutation.

We resolve all the crossings in the tangle using the skein relation: $F(A)$ is the polynomial in A multiplying the resulting tangle of two vertical strands. $G(A)$ is the polynomial coefficient multiplying the tangle of two horizontal strands. Notice that the crossings do not change under rotation. Thus we see that $\langle D \rangle$ and its mutant $\langle D' \rangle$ are equal.

$$\begin{aligned}
& \left\langle \begin{array}{c} \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \end{array} \right\rangle = \langle D \rangle = \langle D' \rangle = \left\langle \begin{array}{c} \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \end{array} \right\rangle \\
& = F(A) \left\langle \begin{array}{c} \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \end{array} \right\rangle + G(A) \left\langle \begin{array}{c} \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \end{array} \right\rangle
\end{aligned}$$

Figure 20. $\langle D \rangle$ and its mutant $\langle D' \rangle$ are equal.

Before we show how writhe is unaffected by mutation, first we want to note that doing a rotation along the horizontal axis then a rotation along the vertical axis will produce a 180° rotation on the diagram. One can show this using the explicit formula for the rotation maps $R_x(\theta)$, $R_y(\theta)$, and $R_z(\theta)$, the rotation along the x -axis, y -axis, and z -axis respectively, where:

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix},$$

$$R_y(\theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix}, \text{ and } R_z(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Thus we have that $R_y(180^\circ) \circ R_x(180^\circ) = R_z(180^\circ)$, i.e. a horizontal rotation then vertical rotation will give us a 180° rotation.

Using this fact, we can just prove that the writhe is unaffected by these two mutation moves. Thus we now include orientation and use writhe to completely show that the Jones polynomial cannot distinguish mutant knots. Let $\langle D \rangle$ be an oriented diagram, and we want to observe how the two mutation moves affect the crossing's orientation in $\langle D \rangle$. In Figure 21, by focusing on the top left oriented strand, we can obtain a total of 6 different arrangements where we start with a strand exiting $\langle D \rangle$. And in Figure 22 we can get yet another 6 different arrangements when we allow the top left string to enter first instead, for a grand total of 12 possible arrangements for how the strings enter and exit $\langle D \rangle$. Now by examining the diagrams in both Figure 21 and Figure 22, we can see that, depending on the figure, applying either a horizontal

rotation or vertical rotation to the local region for each of the diagrams will either preserve the orientation of the tangle inside of the circle A , or reverse the orientation.

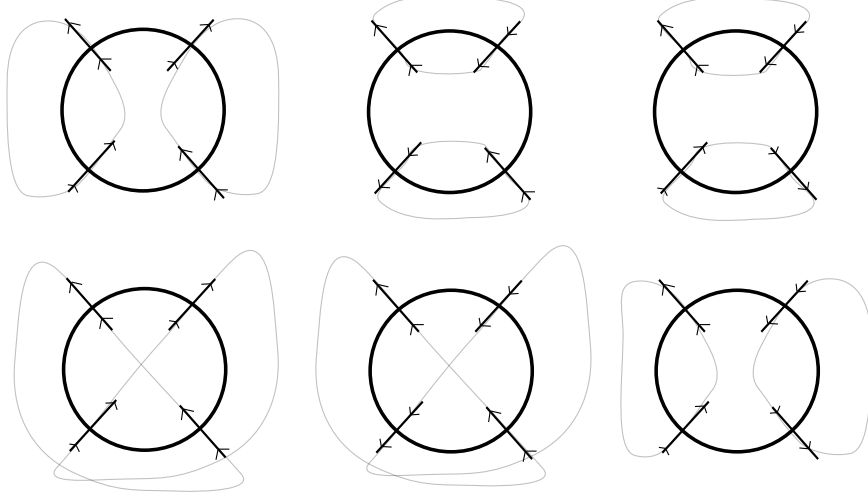


Figure 21. Possible Combinations of the Oriented String Exiting First in $\langle D \rangle$.

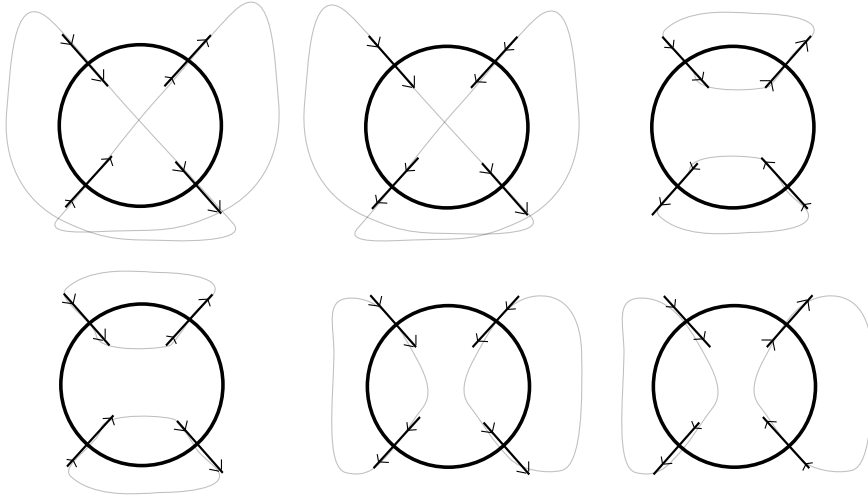


Figure 22. Possible Combinations of the Oriented String Entering First in $\langle D \rangle$.

If the mutation move preserves the orientation inherited from the exterior of the tangle, then the rest of the knot's orientation outside of the mutation circle will also remain the same; meaning the writhe will be unchanged. And if the mutation move reverses the orientation, then we would need to reverse the orientation of the strands throughout the rest of the knot diagram, i.e. the strands outside of the mutation circle containing T . In doing so, this allows the rest of the diagram outside the circle to have consistent orientations. Note that reversing the orientation of any crossings preserves the writhe. In other words, writhe is unchanged under reversed orientation. Thus we can conclude that both of these rotations do not affect the writhe, i.e. the writhe is preserved under mutation. $\square$

Because the Jones polynomial is unable to detect the mutant knot, this gives us motivation to find a new invariant that can perhaps detect mutant knots that are different.

CHAPTER IV

KHOVANOV HOMOLOGY

4.1 A Stronger Knot Invariant

We will now take a look at *Khovanov Homology*, which is regarded as a categorification of the Jones polynomial [16]. For this chapter, we follow Turner's exposition of this invariant [23], with some modifications. Khovanov homology was developed in the late 1990s by Mikhail Khovanov. It is an oriented knot invariant that arises as the homology of a chain complex. Given a knot diagram D , Khovanov constructs, in a purely combinatorial way, a bi-graded chain complex $C^{*,*}(D)$, the *Khovanov complex*, associated to D , i.e. $D \rightsquigarrow C^{*,*}(D)$. Khovanov homology is then the set of homology groups of the chain complex, which gives us $KH^{*,*}(D)$. The Jones polynomial [13] is the graded Euler characteristic of the Khovanov complex, both can be defined by means of the Kauffman bracket [14]. And in this sense, Khovanov homology is a categorification of the Jones polynomial. On the other hand, Khovanov homology is a stronger invariant than the Jones polynomial because there are knots which have the same Jones polynomial, but different Khovanov homologies, such as the 5_1 knot and the 10_{132} knot [4].

Definition 4.1. A *chain complex* is a sequence of abelian groups, C^p , and abelian group homomorphisms (also known as differentials), d^p , such that:

$$\dots \xrightarrow{d^{p+1}} C^p \xrightarrow{d^p} C^{p-1} \xrightarrow{d^{p-1}} \dots$$

and $d^p \circ d^{p+1} = 0$ for each p .

Definition 4.2. A *cochain complex* is the dual notion of a chain complex, such that:

$$\dots \xrightarrow{d^{p-1}} C^{p-1} \xrightarrow{d^p} C^p \xrightarrow{d^{p+1}} \dots$$

and $d^{p+1} \circ d^p = 0$ for each p .

Definition 4.3. The p^{th} *homology group* of a chain complex is $H^p(C) := \frac{\ker(d^p)}{\text{im}(d^{p+1})}$ i.e. H^p is the set of cycles, $\ker(d^p)$, modulo boundaries, $\text{im}(d^{p+1})$, in degree p .

Definition 4.4. The p^{th} *cohomology group* of a cochain complex is $H^p(C) := \frac{\ker(d^p)}{\text{im}(d^{p+1})}$ i.e. H^k is the set of co-cycles, $\ker(d^p)$, modulo co-boundaries, $\text{im}(d^{p+1})$, in degree p .

Remark 4.5. For Khovanov homology we are actually finding co-homology groups, rather than homology groups; thus, we are really doing Khovanov *cohomology*. However, to align with current literature, we will just say Khovanov homology, when we really mean Khovanov cohomology. As such, all mentions of chain complexes are indeed cochain complexes.

4.2 The Set Up

In this section, we will explore the construction of Khovanov homology by first exploring the similarities between the construction of the Kauffman bracket's state sum and the construction of the chain complex for Khovanov's invariant, with an emphasis on a hyper-cube construction. After we have constructed the hyper-cube, we then discuss how it will help us create our sequence of chain complexes for the knot diagram. In order to illustrate all of this, we will work with the trefoil knot. The completed example with the computations of the trefoil knot has been done in Bar-Natan's paper [4].

To begin it is worth recalling the construction of the Kauffman bracket from Chapter 3. Furthermore there are a few minor changes in regards to notation as well. Recall in Definition 3.1 that the Kauffman bracket is based on the α and β smoothings. With a slight change in definition and notation, we will now call the α smoothing the 0 smoothing, and the β smoothing the 1 smoothing; however, neither smoothings will have weights A and A^{-1} respectively here.

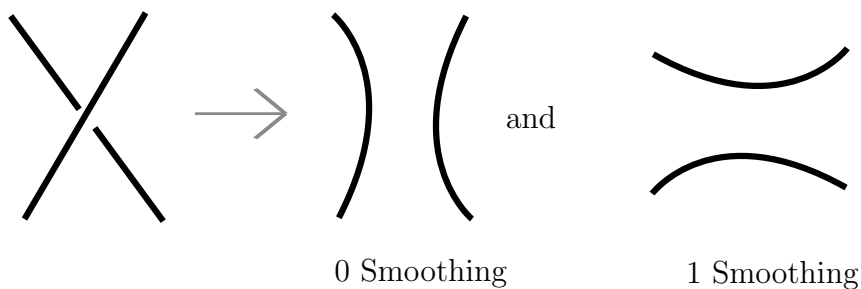


Figure 23. The 0 and 1 Smoothings.

The notation for the number of crossings and both positive and negative crossings will stay the same. Now notice that when we apply our two possible smoothings to a knot with n crossings, we will have 2^n total states for the knot diagram.

To now apply our smoothings in a systematic way, we first number the crossings of the knot diagram with numbers $1, 2, 3, \dots, n$, i.e., the number of crossings the diagram contains. We then index it by a word of n 0's and 1's, i.e. an element of $\{0, 1\}^n$. This will indicate where we applied the 0 and 1 smoothings according to the crossings on our given knot diagram.

Example 4.6. Given a knot diagram with some number of crossings, say c , then we will have 2^c total smoothings, and the set $\{0, 1\}^c$ is the vertex set of a hyper-cube of words, with an edge between words differing in exactly one place. We can think of

each of the words as living on its appropriate vertices of the cube as seen in Figure 24.

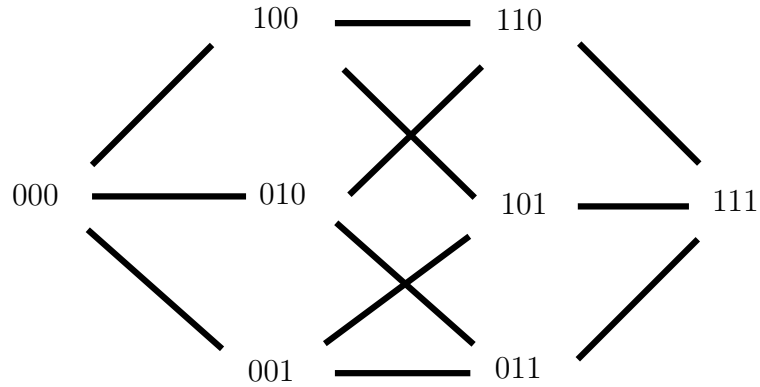


Figure 24. Hyper-Cube of Words for 3-Crossing Knot.

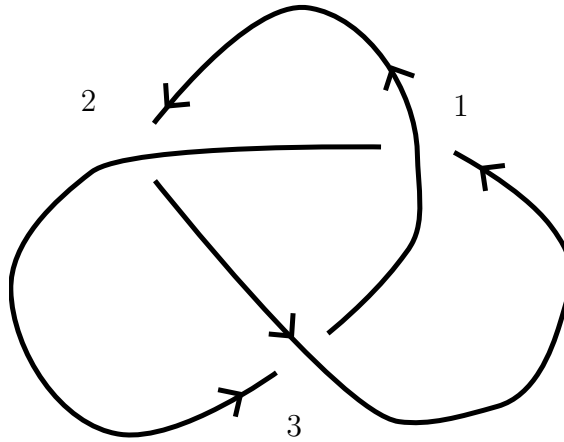


Figure 25. Labeled Trefoil Knot.

Definition 4.7. We turn each edge of the hyper-cube into a *right arrow*, ξ , such that for the position that changes in our word $\alpha \in \{0, 1\}^n$, we label a ‘ $\star$ ’ at that position. In other words, a right arrow ξ connects $\alpha \xrightarrow{\xi} \alpha'$, where $\alpha, \alpha' \in \{0, 1\}^n$. For example,

in Figure 26, the right arrow $\xi = \star 00$, from edge labeled 000 to 100.

Remark 4.8. Note that we will sometimes omit labeling ξ from the rest of the figures following Figure 26 to avoid cluttering the diagrams.

To further expand our hyper-cube of words, we can include a picture of each state, which also live on the vertices of the cube, in which each crossing has been resolved via the 0 and 1 smoothings. We apply the smoothings to the crossings based on Figure 25.

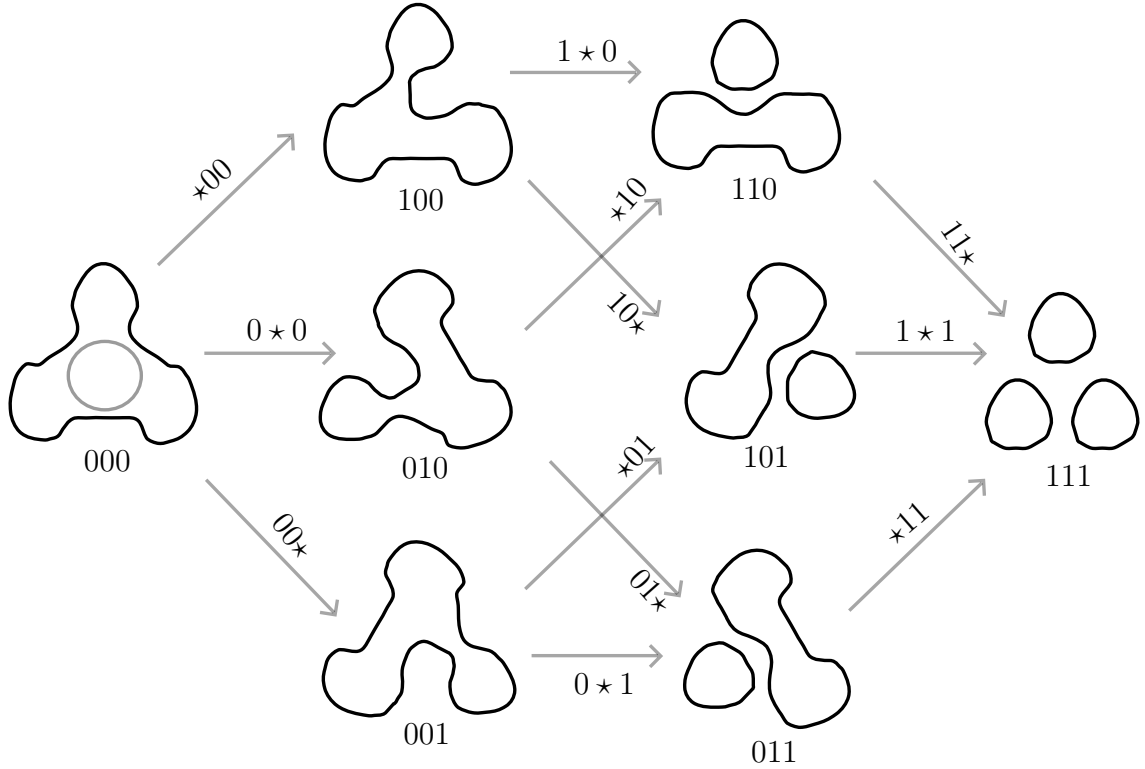


Figure 26. Trefoil Cube of Words with Associated States.

We will now be working over the field $\mathbb{Q}$. Let $V = \mathbb{Q}\{1, x\}$ denote a graded vector space with basis 1 and x . We assign degrees to these basis, so $\deg(1) = 1$ and

$$\deg(x) = -1.$$

Definition 4.9. For an element $\alpha \in \{0, 1\}^n$, we denote r_α = the number of 1's in α , which we also call the height.

For example, in Figure 27, the r_α at $V_{100}, V_{010}, V_{001}$ (which is $C^{1,*}$ in Figure 28) is equal to 1, as we see this height has only one 1 in α .

Definition 4.10. For an element $\alpha \in \{0, 1\}^n$, we denote k_α = the number of circles in the associated state, i.e. the collection of circles in the plane of the specified state.

For example, in Figure 26, the k_α at states 000 and 111 is equal to 2 and 3 respectively, since there are 2 and 3 circles in the plane at these states.

Now to begin constructing our chain complex, we have that for some element $v \in V_\alpha \subset C^{*,*}(D)$, where $v \in C^{i,j}(D)$ has homological grading i and q -grading j (this is formally defined in Definition 4.11). At each edge ξ , we replace each $\alpha \in \{0, 1\}^n$ with the graded vector space V_α .

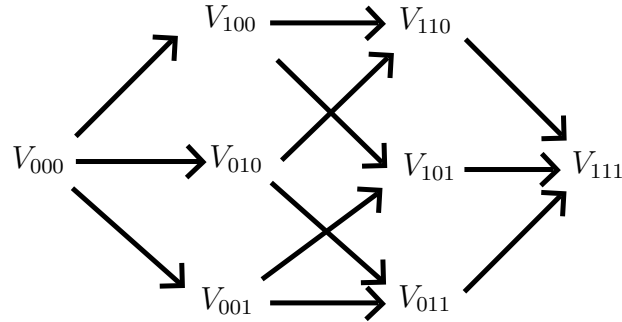


Figure 27. Cube of Words with Graded Vector Space V_α for 3-crossing Knot.

Definition 4.11. For our chain complex $C^{i,j}$, the *homological grading* $i = r_\alpha - n_-$, and the *q-grading* $j = \deg(v) + i + n_+ - n_-$.

Definition 4.12. We can construct our chain complex $C^{i,*}(D)$, where

$$C^{i,*}(D) = \bigoplus_{r_\alpha = i+n_-} V_\alpha$$

We have that $V_\alpha = V^{\otimes k_\alpha}$ (the tensor product of k_α copies of V).

In other words, $C^{i,*}(D)$ is the direct sum of vector spaces for V_α at each height r_α . A way to represent this is by constructing the *cube of resolutions*.

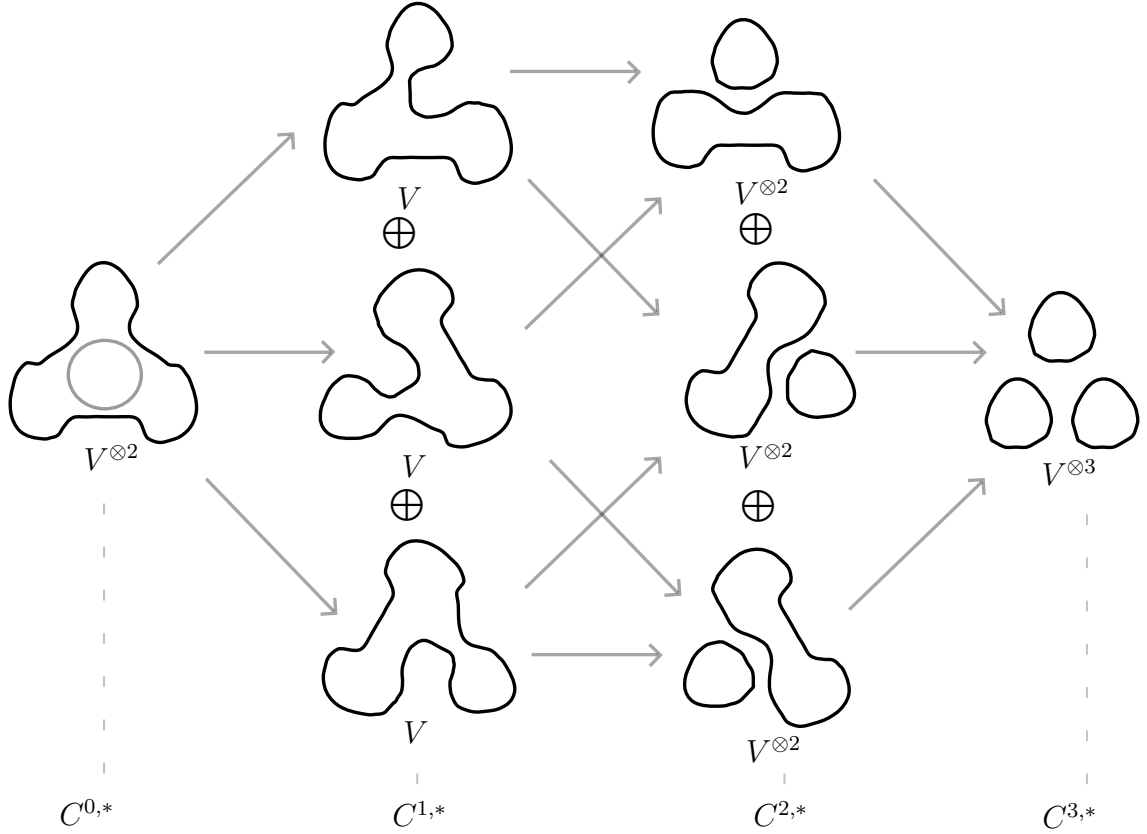


Figure 28. Cube of Resolution for Trefoil with States.

Remark 4.13. Notice that our i -grading in Figure 28, is from the fact that Figure 25

has 3 positive crossings and 0 negative crossings.

At each edge, we have that either two circles merge together or a circle splits into two. Thus, we attach a linear map $d_\xi : V_\alpha \rightarrow V_{\alpha'}$, where $\alpha \xrightarrow{\xi} \alpha'$ connects. Because each circle in a state has a copy of vector space V , we define two linear maps based on this merging and splitting, with respect to the basis vectors 1 and x . We will denote the map m that merges and the map Δ that splits.

Definition 4.14.

$$m : V \otimes V \rightarrow V, \text{ where } m : \begin{cases} 1 \otimes 1 \rightarrow 1 \\ 1 \otimes x \rightarrow x \\ x \otimes 1 \rightarrow x \\ x \otimes x \rightarrow 0 \end{cases}$$

$$\Delta : V \rightarrow V \otimes V, \text{ where } \Delta : \begin{cases} 1 \rightarrow 1 \otimes x + x \otimes 1 \\ x \rightarrow x \otimes x \end{cases}$$

We can now define our differential, $d^q : C^{i,*}(D) \rightarrow C^{i+1,*}(D)$.

Definition 4.15. So for $v \in V_\alpha \subset C^{*,*}(D)$, we set

$$d^q(v) := \sum_{\xi: \text{tail}(\xi)=\alpha} \text{sign}(\xi) d_\xi(v)$$

where $\text{sign}(\xi) = (-1)^{\text{number of 1s to the left of } \star \text{ in } \xi}$

Lemma 4.16. $d^{q+1} \circ d^q = 0$

Sketch of proof: Using the fact that our cube of smoothings is anti-commutative

(when taking into consideration the sign), when we take $\Delta \circ m$ and $m \circ \Delta$, everything will cancel out nicely and give us 0.

We have all we need now to setup an entire chain complex for any knot we want to work with, and thus can find its Khovanov homology.

4.3 Example Computation With Khovanov Homology

Before taking a look at our pair of mutant knots, we will discuss a fully worked out example of applying this invariant to tie everything together. We will be computing the figure-eight knot.

Example 4.17. We will be using this diagram of the figure-eight knot with the associated orientation.

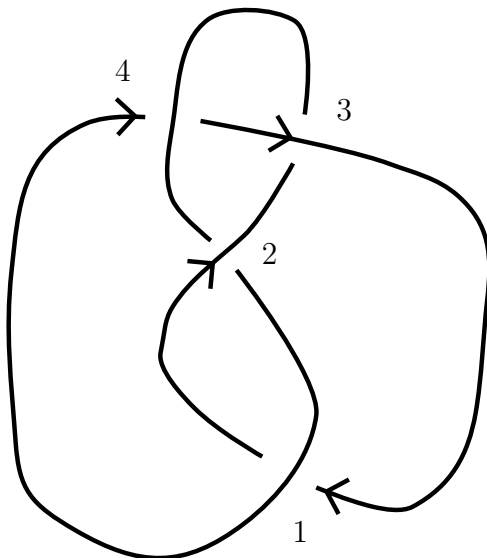


Figure 29. Figure-eight Knot Khovanov Homology Example.

We then construct our cube of words with associated states for the knot.

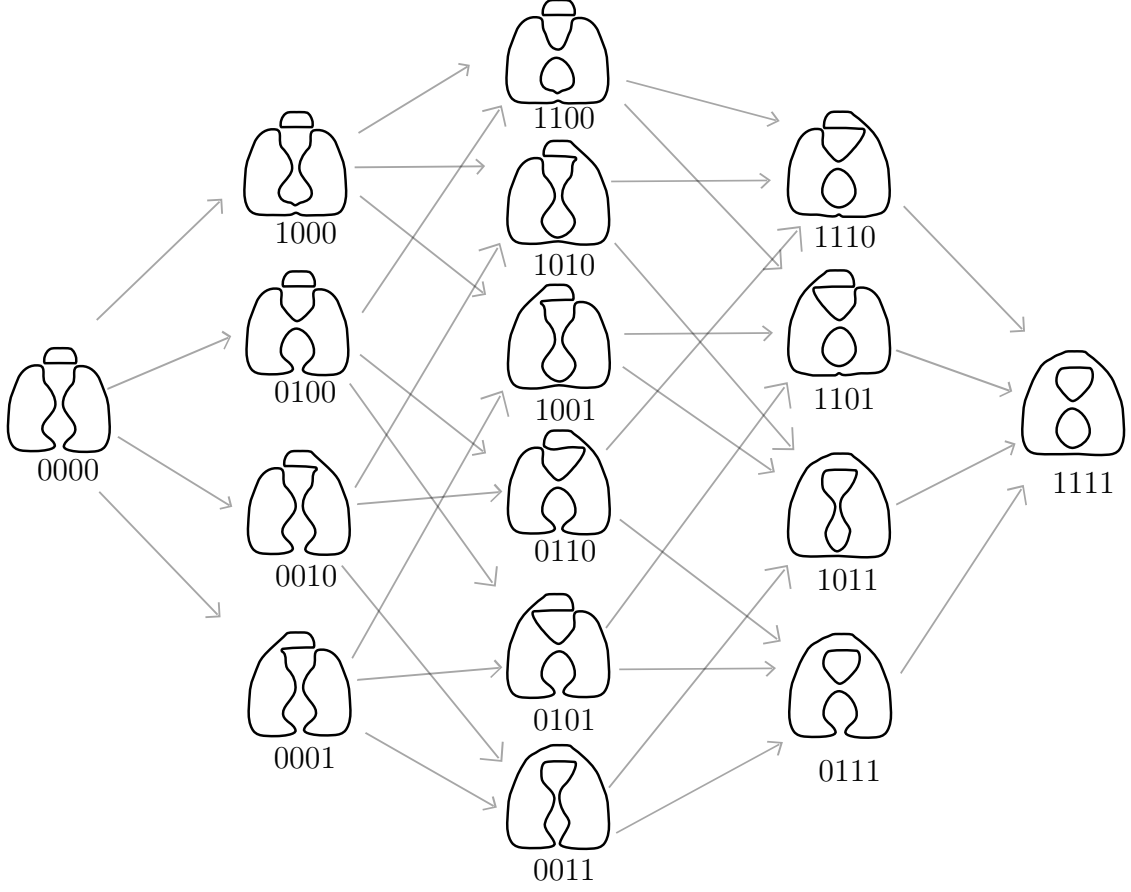


Figure 30. Figure-eight Cube of Words with States.

From our cube of words with their associates states we can then construct our cube of resolutions. Recall that at each height, we are taking the direct sum of vector spaces for $V^{\otimes k_\alpha}$. Furthermore, we have also labeled each edge with its appropriate merging or splitting map, including its appropriate sign.

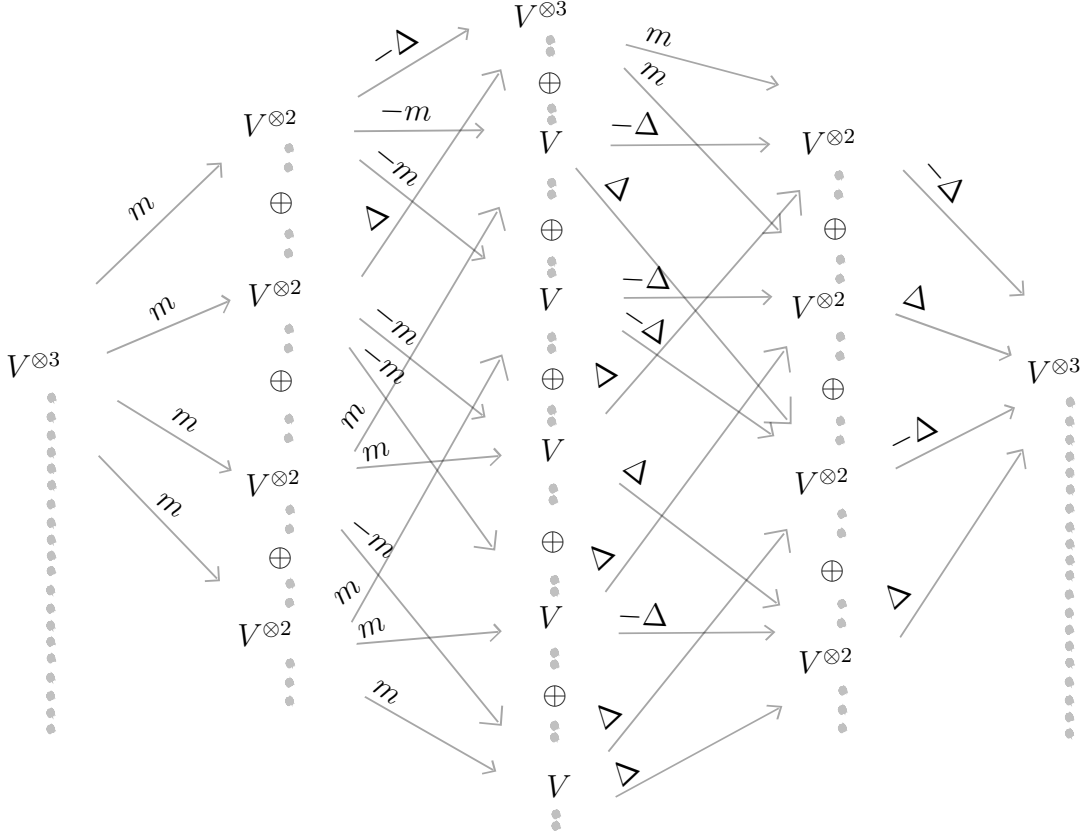


Figure 31. Figure-eight Cube of Smoothing.

Further we can construct our chain complex as follows:

$$0 \rightarrow C^{-2,*} \xrightarrow{d^{-3}} C^{-1,*} \xrightarrow{d^{-2}} C^{0,*} \xrightarrow{d^{-1}} C^{1,*} \xrightarrow{d^0} C^{2,*} \rightarrow 0$$

Using the labeling as seen in Figure 32, we can define our differential d^{-3} as follows:

$$d^{-3}(v_1 \otimes v_2 \otimes v_3) = (m(v_1 \otimes v_3) \otimes v_2, m(v_1 \otimes v_3) \otimes v_2, m(v_1 \otimes v_2) \otimes v_3, m(v_2 \otimes v_3) \otimes v_1)$$

So we have for d^{-3} :

$$1 \otimes 1 \otimes 1 \mapsto (1 \otimes 1, 1 \otimes 1, 1 \otimes 1, 1 \otimes 1)$$

$$x \otimes 1 \otimes 1 \mapsto (x \otimes 1, x \otimes 1, x \otimes 1, 1 \otimes x)$$

$$1 \otimes x \otimes 1 \mapsto (1 \otimes x, 1 \otimes x, x \otimes 1, 1 \otimes x)$$

$$1 \otimes 1 \otimes x \mapsto (x \otimes 1, x \otimes 1, 1 \otimes x, x \otimes 1)$$

$$x \otimes x \otimes 1 \mapsto (x \otimes x, x \otimes x, 0, x \otimes x)$$

$$x \otimes 1 \otimes x \mapsto (0, 0, x \otimes x, x \otimes x)$$

$$1 \otimes x \otimes x \mapsto (x \otimes x, x \otimes x, x \otimes x, 0)$$

$$x \otimes x \otimes x \mapsto (0, 0, 0, 0)$$

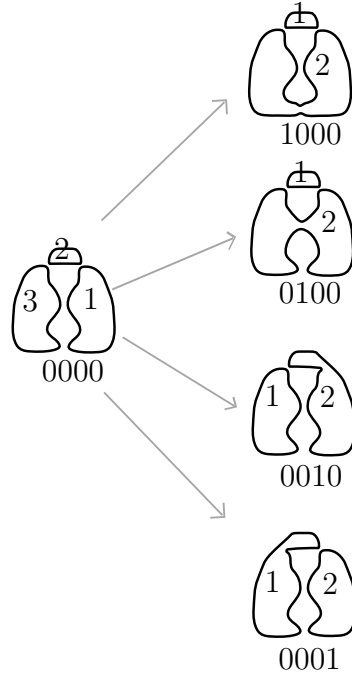


Figure 32. Labeling for d^{-3} .

Because of the many states involved with the knot, it would be easier to construct a matrix representation of the ordered basis, where each basis vector is mapped via our linear map from one height to the next. We can then use this matrix to determine the cycles and boundaries at each differential. We construct this matrix where we have the basis of the domain (from left to right: $1 \otimes 1 \otimes 1, x \otimes 1 \otimes 1, 1 \otimes x \otimes 1, 1 \otimes$

$1 \otimes x, x \otimes x \otimes 1, x \otimes 1 \otimes x, 1 \otimes x \otimes x, x \otimes x \otimes x$) and the basis of the image (from top to bottom: $(1 \otimes 1, 0, 0, 0), \dots, (0, 0, 0, 1 \otimes 1), (x \otimes 1, 0, 0, 0), \dots, (0, 0, 0, x \otimes 1), (1 \otimes x, 0, 0, 0), \dots, (0, 0, 0, 1 \otimes x), (x \otimes x, 0, 0, 0), \dots, (0, 0, 0, x \otimes x)$). So for d^{-3} we can construct a matrix as follows:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

After some calculations, we can determine that for d^{-3} , we have that a basis for the nullspace (corresponding to $\ker(d^{-3})$) of this matrix is:

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

From this we can determine that the cycles are $x \otimes x \otimes x$. Now the column space for this matrix corresponds to $\text{img}(d^{-3})$, and we have that its corresponding basis are:

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

So we have that the homology group at height 0 is: $H^{-2} = \frac{\ker(d^{-3})}{\text{img}(d^{-4})}$ where the cycle is: $x \otimes x \otimes x$, and the boundary is empty here, thus the homology group is isomorphic to $\mathbb{Q}$.

Now using the labeling of circles, as seen in Figure 33, we can define our differential d^{-2} as follows:

$$\begin{aligned} d^{-2}(v_1 \otimes v_2, v_3 \otimes v_4, v_5 \otimes v_6, v_7 \otimes v_8) = \\ (v_1 \otimes -\Delta(v_2) + v_3 \otimes \Delta(v_4), m(v_5 \otimes v_6) - m(v_1 \otimes v_2), m(v_7 \otimes v_8) - m(v_1 \otimes v_2), \\ m(v_5 \otimes v_6) - m(v_3 \otimes v_4), m(v_7 \otimes v_8) - m(v_3 \otimes v_4), m(v_7 \otimes v_8) - m(v_5 \otimes v_6)) \end{aligned}$$

So we have for d^{-2} :

$$\begin{aligned} (1 \otimes 1, 0, 0, 0) &\mapsto (1 \otimes -(1 \otimes x + x \otimes 1), -1, -1, 0, 0, 0) \\ (0, 1 \otimes 1, 0, 0) &\mapsto (1 \otimes (1 \otimes x + x \otimes 1), 0, 0, -1, -1, 0) \\ (0, 0, 1 \otimes 1, 0) &\mapsto (0, 1, 0, 1, 0, -1) \\ (0, 0, 0, 1 \otimes 1) &\mapsto (0, 0, 1, 0, 1, 1) \\ (x \otimes 1, 0, 0, 0) &\mapsto (x \otimes -(1 \otimes x + x \otimes 1), -x, -x, 0, 0, 0) \\ (0, x \otimes 1, 0, 0) &\mapsto (x \otimes (1 \otimes x + x \otimes 1), 0, 0, -x, -x, 0) \\ (0, 0, x \otimes 1, 0) &\mapsto (0, x, 0, x, 0, -x) \\ (0, 0, 0, x \otimes 1) &\mapsto (0, 0, x, 0, x, x) \\ (1 \otimes x, 0, 0, 0) &\mapsto (1 \otimes -(x \otimes x), -x, -x, 0, 0, 0) \\ (0, 1 \otimes x, 0, 0) &\mapsto (1 \otimes (x \otimes x), 0, 0, -x, -x, 0) \\ (0, 0, 1 \otimes x, 0) &\mapsto (0, x, 0, x, 0, -x) \\ (0, 0, 0, 1 \otimes x) &\mapsto (0, 0, x, 0, x, x) \\ (x \otimes x, 0, 0, 0) &\mapsto (x \otimes -(x \otimes x), 0, 0, 0, 0, 0) \\ (0, x \otimes x, 0, 0) &\mapsto (x \otimes (x \otimes x), 0, 0, 0, 0, 0) \\ (0, 0, x \otimes x, 0) &\mapsto (0, 0, 0, 0, 0, 0) \end{aligned}$$

$$(0, 0, 0, x \otimes x) \mapsto (0, 0, 0, 0, 0, 0)$$

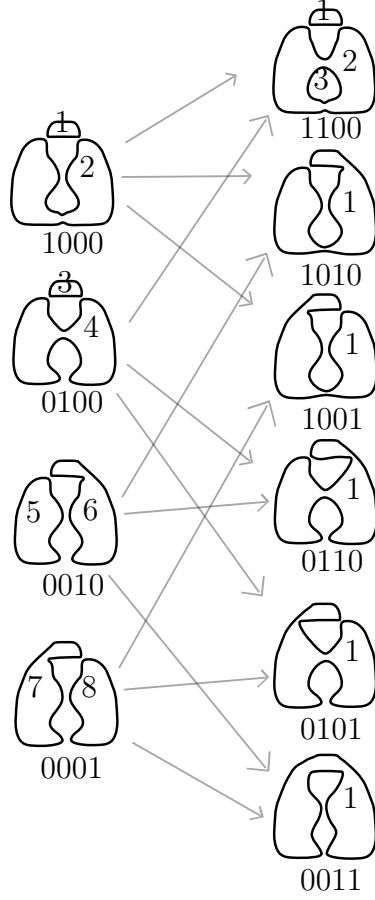


Figure 33. Labeling for d^{-2} .

Using the same idea as before, we can create another matrix to find the kernel and image of d^{-2} . This time the columns of the matrix are now the boundaries of d^{-3} , and its rows are (from top to bottom): $(1, 0, 0, 0, 0, 0)$, $(0, 1, 0, 0, 0, 0)$, $(0, 0, 1, 0, 0, 0)$, $(0, 0, 0, 1, 0, 0)$, $(0, 0, 0, 0, 1, 0)$, $(0, 0, 0, 0, 0, 1)$, $(x, 0, 0, 0, 0, 0)$, $(0, x, 0, 0, 0, 0)$, $(0, 0, x, 0, 0, 0)$, $(0, 0, 0, x, 0, 0)$, $(0, 0, 0, 0, x, 0)$, $(0, 0, 0, 0, 0, x)$, $(1 \otimes 1 \otimes 1, 0, 0, 0, 0, 0)$, $(x \otimes 1 \otimes 1, 0, 0, 0, 0, 0)$, $(1 \otimes x \otimes 1, 0, 0, 0, 0, 0)$, $(x \otimes x \otimes 1, 0, 0, 0, 0, 0)$,

$(x \otimes 1 \otimes x, 0, 0, 0, 0, 0), (1 \otimes x \otimes x, 0, 0, 0, 0, 0), (x \otimes x \otimes x, 0, 0, 0, 0, 0)$. So for d^{-2} we can construct a matrix as follows:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 \end{bmatrix}$$

After some calculations, we can determine that the basis for the nullspace of this matrix are:

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},
\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},
\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ -1 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},
\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},
\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},
\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},
\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

At height 1, after some calculations, we can determine that the homology group for H^{-1} is: $\langle (0, 0, 1 \otimes x - x \otimes 1, 0) \rangle$, thus the homology group is isomorphic to $\mathbb{Q}$.

We can apply the same exact procedure for calculating d^{-1} and d^0 . The rest of the results for the homology of this knot can be found in the knot atlas [17].

4.4 Applying This to Our Pair of Knots

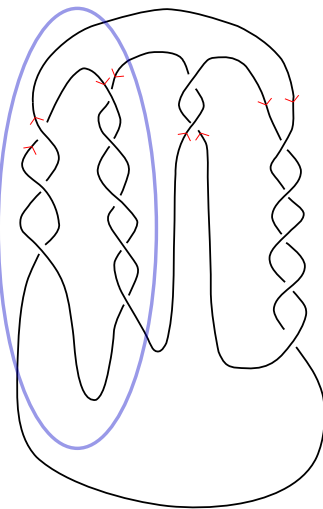
Since we have now established how to perform the computation for Khovanav's invariant, we want to start taking a closer look at the motivating idea behind this thesis,

namely mutation. Our research question is as follows:

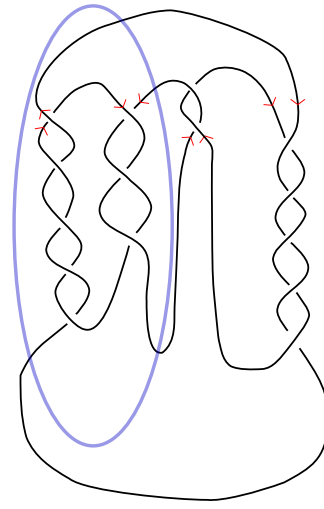
Question 4.18. Is the Khovanov homology of a knot invariant under mutation?

In recent literature regarding Khovanov homology and mutation, Bloom [5] and Wehrli [24], [25] have made extensive headway in this area. We say Khovanov homology can detect mutation if there is a pair of mutant knots for which their Khovanov homologies are distinct; however, this would only make sense if we talk about knots that are not isotopic to their mutant knots. We decided to look into a specific pretzel knot and a distinct mutant of it, namely the $P(-3, -5, 2, 5)$ and $P(-5, -3, 2, 5)$ pretzel knots.

Definition 4.19. A pretzel knot $P(a_1, a_2, \dots, a_k)$ is a knot with k columns, if and only if $k \geq 0$, all of the a_i 's and k are odd or $k \geq 1$ and only one of the a_i is even [15].



$P(-3, -5, 2, 5)$



$P(-5, -3, 2, 5)$

Figure 34. Pretzel knots $P(-3, -5, 2, 5)$ and $P(-5, -3, 2, 5)$.

Remark 4.20. The purple circle in Figure 34 indicates where the mutation occurs between these two pretzel knots.

Definition 4.21. Given a knot $K \in \mathbb{S}^3$, the *knot exterior*, $E(K) = \mathbb{S}^3 - \text{int}N(K)$, is the 3-manifold that is left after removing a tubular neighborhood, $N(K)$, of the knot in $\mathbb{S}^3$. In other words, it is the complement of an open solid torus knotted like the knot, of which the solid we remove is the tubular neighborhood of the knot [1].

Definition 4.22. A properly embedded surface in a 3-manifold is called *essential* if its fundamental group injects into the fundamental group of its knot exterior.

Definition 4.23. A number $\frac{p}{q} \in \mathbb{Q} \cup \{\infty\}$ is a *slope* of a properly embedded surface in the knot exterior, $E(K)$, with a boundary component representing $p[\mu] + q[\lambda] \in H_1(\partial E(K))$ with respect to the standard meridian μ and longitude λ . A slope $\frac{p}{q}$ is a *boundary slope* if there is an essential surface with slope $\frac{p}{q}$. Then we define the set of boundary slopes of K to be: $bs(K) = \{r \in \mathbb{Q} \cup \{\infty\} | r \text{ is a boundary slope of } K\}$ [3].

The set of boundary slopes is an invariant of a knot. It is a finite set as shown by Hatcher [11].

Definition 4.24. Recall from Definition 3.3, a Kauffman state of a knot diagram D will give rise to a collection of circles that are embedded in our projection plane $\mathbb{S}^2$. Each of the circles bound a disk in a ball below $\mathbb{S}^2$, such that the collection of these disks is unique up to isotopy. The *state surface* is where the surface is formed from the collection of disks which are joined by half-twisted bands. Each half-twisted band is created for each crossing where a resolution was chosen [10].

Remark 4.25. Pretzel knots $P(-3, -5, 2, 5)$ and $P(-5, -3, 2, 5)$ are indeed distinct mutant knots because the set of boundary slopes of these pretzel knots are different

[12]. For the $P(-3, -5, 2, 5)$ knot, we have that the set of boundary slopes are: $[-16, -10, -8, -6, -4, -2, 0, 2, 3, 4, 8, 14, \infty]$. And for the $P(-5, -3, 2, 5)$, we have that the set of boundary slopes are: $[-16, -10, -8, -6, -4, -2, 0, 3, 6, 8, 14, \infty]$.

Now because these knots both would have 2^{15} possible smoothings to look at, our method was to look at a specific smoothing for each of these knots, namely the 000111111100000 smoothing, for the knot on the left and the 111110001100000 smoothing for the knot on the right in Figure 34. The state surfaces for these states surfaces both have slope 6 [9], but one of them is essential for $P(-5, -3, 2, 5)$, and the other one is not essential. It is natural to ask whether the Khovanov homology classes differ for these generators of each state.

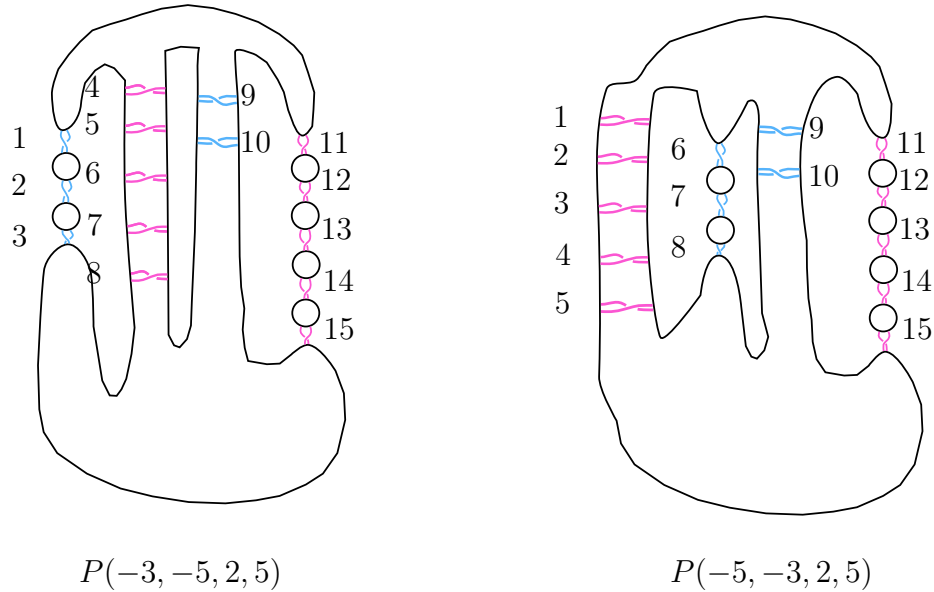


Figure 35. State Surface Diagrams (Blue: 0 smoothing, Red: 1 smoothing). The number next to each crossing, indicates the numbering.

We let $x \otimes x \otimes x \otimes x \otimes x \otimes x \otimes x \otimes x$ be the generator, where all the circles are marked x , for the 000111111100000 and 111110001100000 Kauffman states respectively. We now have our lemma:

Lemma 4.26. *$x \otimes^8 x$ is a cycle for both pretzel knots.*

Proof. Looking at the particular states for both of the pretzel knots, we can see that we have 8 circles in Figure 35. The boundary maps of both of these states are all merge maps. Thus the generator, which comes from assigning each of these circles the x basis, would go to 0 under the boundary map. $\square$

CHAPTER V

CONCLUSIONS

So overall we have gone through some basic necessities of knot theory to be able to discuss knot mutations and the Jones polynomial, and to be able to motivate our particular problem. Then by examining, via Khovanov homology, our pretzel knot and its mutant, we have so far been able to find a cycle for both of these knots, namely the cycle where all the elements in the state circles contain x . The next step we are looking into is to determine whether or not it is a boundary as well for both these knots. If we can determine that it is a boundary for one knot but not the other, we could conclude that Khovanov homology does detect mutation of knots.

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BIOGRAPHICAL SKETCH

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